

DM
Assignment - 02

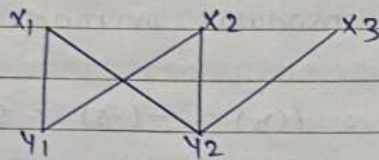
DATE: 01/10/23

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• PART-A ($5 \times 2 = 10$)

Q1. Define the Bipartite graph.

Ans A bipartite graph is a graph in which the vertices can be divided into two disjoint sets, such that no two vertices within the same set are adjacent. In other words, it is a graph in which every edge connects a vertex of one set to a vertex of the other set.



Q2. How many edges are there in a graph with 7 vertices each of degree 4?

Ans

$$\text{Total no. of edges} = \frac{(\text{No. of vertices}) \times (\text{Degree of each vertex})}{2}$$

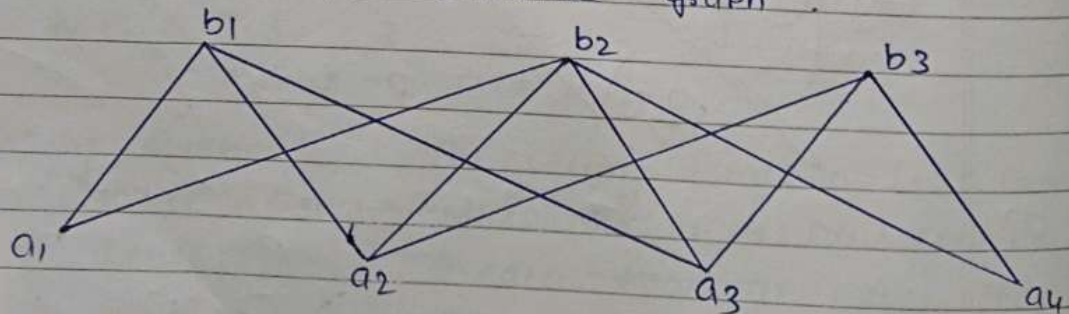
$$= \frac{7 \times 4}{2} = \frac{28}{2} = 14$$

$$\text{Total no. of edges} = 14.$$

Q3. Write an example of a complete Bipartite graph.

Ans.

Example of Complete Bipartite graph :



Q4.

Ans

Define Tree and forest.

Tree: in graph Theory, a tree is undirected, connected and acyclic graph. In other words, a connected graph that does not contain even a single cycle is called tree.

A connected graph with no cycles is called a tree. A tree with n vertices has $(n-1)$ edges.

ex:

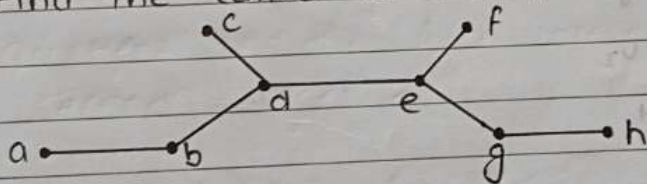
Forested



- Forest: in graph Theory, A disconnected acyclic graph is called a forest. In other words, a disjoint collection of trees is called a forest. Each component of a forest is tree.

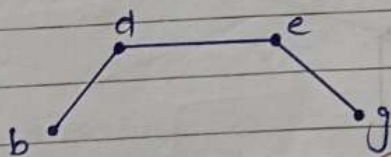
Q5.

Find the centre of the following tree.

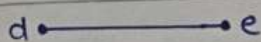


Ans:

Remove all vertices of degree 1 and also remove their incident edges, we get following tree.



Again remove all vertices of degree 1 and also remove their incident edges, we get,

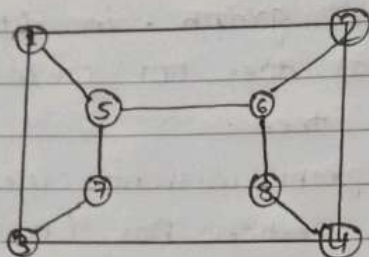


As two vertices joined by an edge is left, this tree has bi-centre 'de' and tree is bi-central.

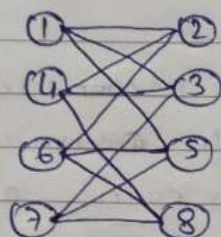
• PART : B (3x7=21)

Q1. Which of the following are Bipartite graph?

(a)

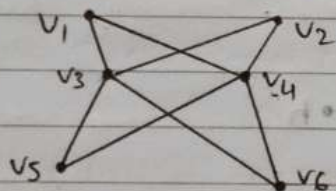


Solⁿ → We can draw above graph in another way.

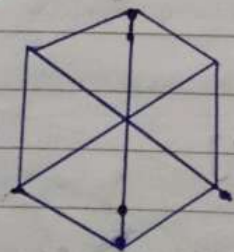


Vertices of same set are not joined, Hence it is Bipartite graph.

(b)

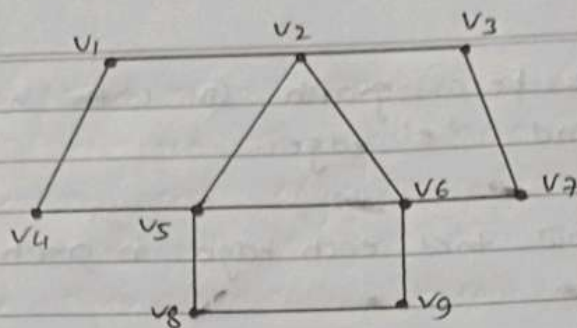
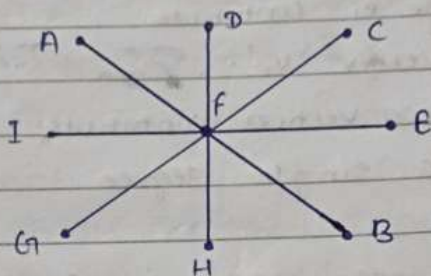


Solⁿ. Draw the another way -



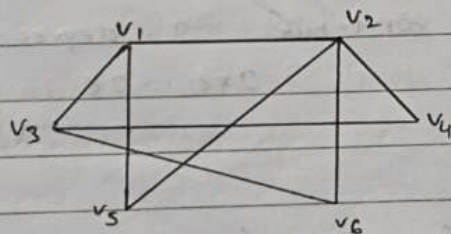
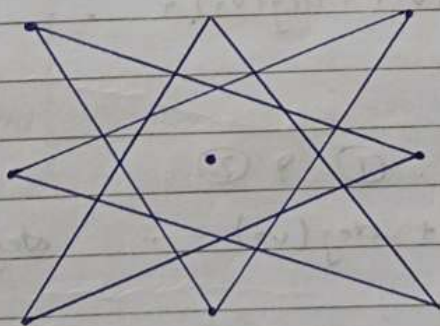
Not a Bipartite Graph.

(C)

solⁿ.

it is a Bipartite Graph.

(d)

solⁿ.

Not a bipartite Graph.

Q2. Prove that if $G = (V, E)$ is a graph with e number of edges, then,

$$\sum_{v \in V} \deg(v) = 2e$$

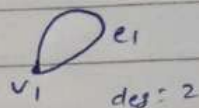
i.e., the sum of degree of the vertices of G is always even.

Solⁿ.

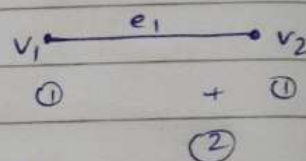
Let there be a graph G with $V = \{v_1, v_2, v_3, \dots, v_n\}$ vertices and ' e ' edges.

As we know that each edge in graph G contribute 2-degree.

(i) loop with edge e_1 contribute 2-degree to vertex ' v_1 '.



(ii) edge with two vertices contributes each vertex a single degree.



So, A single edge contribute 2-degree.

$\therefore$ A single edge contribute 2-degree.

$\therefore$ e no. of edge = $2xe = 2e$

————— (i)

$\therefore$ Total degree of graph G (in terms of vertices)

$$\deg(v_1) + \deg(v_2) + \deg(v_3) + \dots + \deg(v_n)$$

————— (ii)

So,

from eqⁿ: (i) & (ii)

$$\deg(v_1) + \deg(v_2) + \deg(v_3) + \dots + \deg(v_n) = 2e$$

so

$$\sum_{v \in V} \deg(v) = 2e$$

pro

it is also called Handshaking Theorem.

Q3

Prove that the Complete graph K_5 on five Vertices is non-Planar.

Ans

The Complete graph K_5 contains 5 vertices and 10 edges.

Now, for connected Planar graph: $3V - e \geq 6$

$$3V - e \geq 6$$

$V = \text{vertices}$

$$3(5) - 10 \geq 6$$

$e = \text{edges}$

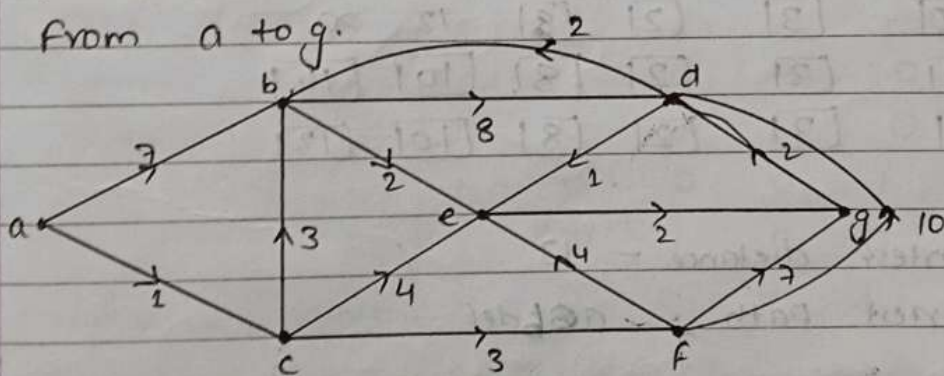
$$5 \geq 6$$

It does not satisfy the Property of Planar graph because it must be greater than or equal to 6. Thus K_5 is a non-planar graph.

• PART-C ($3 \times 7 = 21$)

Q1. Find the shortest path

(a) from a to g.

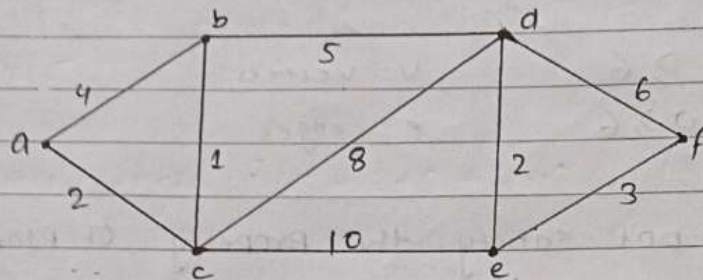


Sol.

a	b	c	d	e	f	g
0	7	1	∞	∞	∞	∞
0	4	1	∞	5	4	∞
0	4	1	12	5	4	∞
0	4	1	12	5	4	11
0	4	1	12	5	4	7
0	4	1	12	5	4	7
0	4	1	9	5	4	7
0	4	1	9	5	4	7

So, Shortest Path distance = 9

(b) From a to f



sol

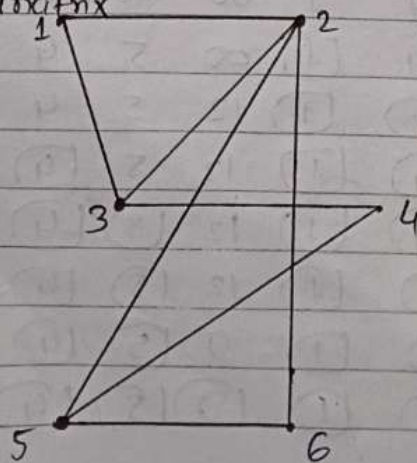
	a	b	c	d	e	f
a	0	∞	∞	∞	∞	∞
b	4	0	2	∞	∞	∞
c	2	3	0	10	12	∞
d	8	3	2	0	12	∞
e	10	3	2	8	0	14
f	14	3	2	8	10	0

Shortest distance - 13

shortest path - a**b**def

Q2. Find the following

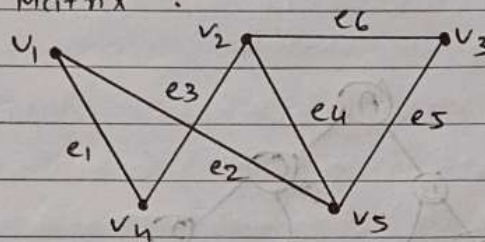
(a) Adjacency matrix



Sol. Adjacency matrix :-

	V_1	V_2	V_3	V_4	V_5	V_6
V_1	0	1	1	0	0	0
V_2	1	0	1	0	1	1
V_3	1	1	0	1	0	0
V_4	0	0	1	0	1	0
V_5	0	1	0	1	0	1
V_6	0	1	0	0	1	0

(b) Incidence Matrix :



Sol. Incidence Matrix :

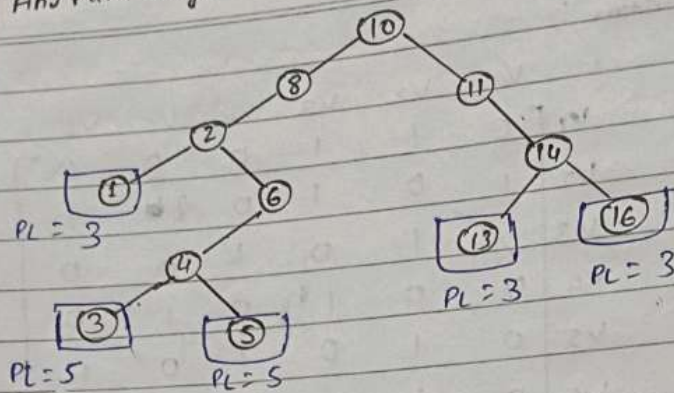
	e_1	e_2	e_3	e_4	e_5	e_6
V_1	1	1	0	0	0	0
V_2	0	0	1	1	0	1
V_3	0	0	0	0	1	1
V_4	1	0	1	0	0	0
V_5	0	1	0	1	1	0

Q3. Define Binary Tree and find the path length of the following binary trees.

Ans in graph theory, A binary tree is a rooted tree in which each node has at most two children, which are referred to as the left child and right child. A node with no children is called a leaf node. A node with one or two children is called an internal node. The top node in the tree is called the root node.
ex: binary trees are used to represent arithmetic expressions, search trees, and file systems.

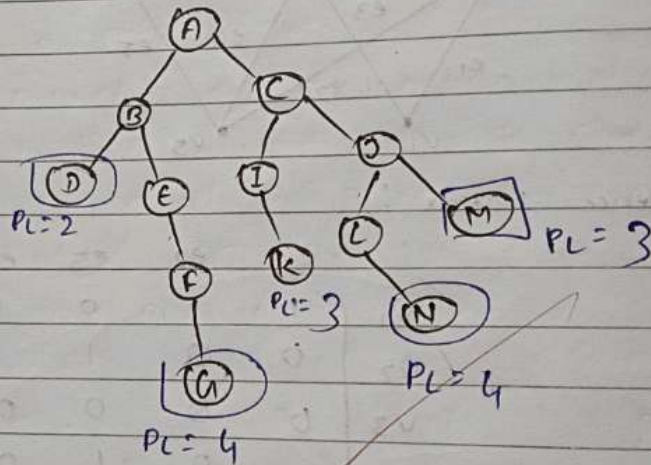
Find Path Length -

(a)



$$\text{Path length} = 3 + 5 + 5 + 3 + 3 = 19$$

(b)



$$\text{Path length} = 2 + 4 + 3 + 4 + 3 = 16$$